

Web Technologies 1 (Frontend)

Week 3: CSS Fundamentals

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1 Overview

Week 3 introduces CSS: what it is, how to add it to a page, selectors and specificity, colors, fonts, the box model, positioning, floats, interactive states, and basic CSS tooling/validation.

2 1. What is CSS?

CSS (Cascading Style Sheets) is the language used to style a web page.

- CSS describes how HTML elements are displayed on screen, paper, or other media.
- CSS saves work: it can control the layout of multiple pages at once.

```
p { color: red; font-size: 10px; }  
/* selector { property: value; property: value; } */
```

3 2. How to Add CSS

There are three ways to insert a style sheet:

- **Inline CSS** — the `style` attribute on an HTML element:

```
<p style="color:blue;">Blue text</p>
```

- **Internal CSS** — a `<style>` element inside `<head>`:

```
<style> p { color: blue; } </style>
```

- **External CSS** — a link to an external `.css` file (lets the same styles be reused across multiple pages):

```
<link rel="stylesheet" href="style.css">
```

4 3. CSS Selectors

Selectors tell CSS which HTML elements to style; the declarations inside `{ }` are then applied to them.

Selector	Syntax	Selects
Element	<code>p</code>	all <code><p></code> elements
Class	<code>.card</code>	elements with <code>class="card"</code>
ID	<code>#title</code>	one element with <code>id="title"</code>

Classes can be reused on multiple elements; IDs should be unique on the page.

5 4. Selector Priority (Specificity)

When multiple selectors target the same element, the more specific one wins:

Element < Class < ID

ID is the most specific (highest priority); an element selector (e.g. `p`, `h1`, `div`) is the least specific.

5.1 Cascade and Inheritance

- The *cascade* resolves declarations that target the same element.
- Specificity compares the *complete* selector, not just its type.
- If specificity is equal, the *later* declaration wins.
- Color and font properties usually inherit from parent to child; margin and border do not.

```
.card p { color: navy; }  
.card .note { color: crimson; }  
/* a <p class="note"> inside .card becomes crimson  
   because the second selector is more specific */
```

5.2 Selector Example

```
<h2 id="title">Menu</h2>  
<p>Coffee</p>  
<p class="off">Tea</p>  
<p class="off">Cake</p>
```

```
p { color: brown; }
.off {
  background: gray;
  color: black;
}
#title { text-align: center; }
```

Result: "Coffee" is brown; "Tea" and "Cake" have a gray background and black text (the more specific `.off` class overrides the element rule); "Menu" is centered.

6 5. Colors in CSS

CSS supports several color formats:

- **Named colors:** red, blue, green
- **HEX:** #ff0000
- **RGB / RGBA:** rgb(255, 0, 0), rgba(255, 0, 0, 0.5)
- **HSL:** hsl(0, 100%, 50%)

```
h1 { color: tomato; }           /* text color */
h1 { background-color: gray; }  /* background color */
h1 { border: 2px solid red; }   /* border color */
```

7 6. Fonts in CSS

- **font-family** — the font (Arial, serif, sans-serif, monospace, custom Google Fonts).
- **font-size** — text size (px, em, rem).
- **font-weight** — thickness (normal, bold, lighter).
- **font-style** — normal or italic.
- **text-align** — left, right, center, justify.

```
p {
  font-family: Arial, sans-serif;
  font-size: 18px;
  font-weight: bold;
  text-align: center;
}
```

8 7. Spacing and the Box Model

Every HTML element is a box with four layers, from inside out:

- **content** — text or other content.
- **padding** — space inside the element (between content and border).
- **border** — outline around the element.
- **margin** — space outside the element.

```
div {  
  padding: 20px;  
  border: 1px solid black;  
  margin: 15px;  
}
```

9 8. Positioning and Alignment

The `position` property controls how an element is positioned:

- **static** — default position (normal document flow).
- **relative** — moved relative to its normal position.
- **absolute** — moved relative to the nearest positioned ancestor.
- **fixed** — stays in the same place relative to the viewport (doesn't scroll away).

`text-align` sets the horizontal alignment of text (`left`, `center`, `right`, `justify`).

10 9. Float and Clear

`float` removes an element from the normal flow and lets content wrap around it.

- `float: left, right, none.`
- `width` is commonly used with floated elements to control their size; floated elements can sit side by side.

`clear` prevents an element from appearing beside floated content (`left`, `right`, `both`, `none`);
`clear: both` places the element below floats on both sides.

```
.box {  
  float: left;  
  width: 33.33%;  
  box-sizing: border-box;  
}
```

11 10. Interactive States

- `:hover` — applies while the pointer is over an element.
- `:focus` — applies when a link or form control receives keyboard focus. A visible focus indicator helps keyboard users track their position.

```
nav a:hover { color: crimson; }
input:focus {
  outline: 3px solid #2457D6;
  outline-offset: 2px;
}
```

12 11. CSS Features

12.1 Favicons

A favicon is a small icon shown in the browser tab next to the page title; usually `.png`, `.ico`, or `.svg`, commonly 16×16 or 32×32 px.

```
<link rel="icon" href="favicon.png">
```

12.2 Sizing Units

- `px` — fixed size.
- `%` — relative to the parent element.
- `em` — relative to the parent element's font size.
- `rem` — relative to the root element's font size.

13 12. CSS Validation and DevTools

- Inspect applied and overridden declarations in the *Styles* panel.
- Use *Computed* to check final values and the box model.
- Test several viewport widths for responsiveness.
- Use keyboard navigation to check focus states.
- Run the W3C CSS Validator (jigsaw.w3.org/css-validator) and fix every error.

14 Summary

Week 3 covers CSS fundamentals: three ways to add styles, element/class/ID selectors and specificity, the cascade and inheritance, colors, fonts, the box model (content/padding/border/margin), positioning (static/relative/absolute/fixed), float/clear, hover/focus states, favicons, sizing units, and CSS validation via DevTools/W3C validator.